AFFINE QUASI-FINITE PUSHFORWARD INDUCES A MORPHISM OF QUOT FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 827-834 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 836-851, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 18 of queue_moduli.json (pb=873, stmt 827-834, proof 836-851).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DN9: sources/stacks-moduli/pinned-original-source.tex, source lines 425–430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07U8: sources/stacks-moduli/pinned-original-source.tex, source lines 851–851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let B be an algebraic space. Let $\pi : X \to Y$ be an affine quasi-finite morphism of algebraic spaces which are separated and of finite presentation over B. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then π_* induces a morphism $\mathrm{Quot}_{\mathcal{F}/X/B} \to \mathrm{Quot}_{\pi_*\mathcal{F}/Y/B}$.*

Proof. Set $\mathcal{G} = \pi_*\mathcal{F}$. Since π is affine we see that for any scheme T over B we have $\mathcal{G}_T = \pi_{T,*}\mathcal{F}_T$ by Cohomology of Spaces, Lemma 07U8. Moreover π_T is affine, hence $\pi_{T,*}$ is exact and transforms quotients into quotients. Observe that a quasi-coherent quotient $\mathcal{F}_T \to \mathcal{Q}$ defines a point of $\mathrm{Quot}_{X/B}$ if and only if $\mathcal{Q}$ defines an object of $\mathcal{C}oh_{X/B}$ over T (similarly for $\mathcal{G}$ and Y). Since we've seen in Lemma

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0DN9 that π_* induces a morphism $\mathcal{C}oh_{X/B} \to \mathcal{C}oh_{Y/B}$ we see that if $\mathcal{F}_T \to \mathcal{Q}$ is in $\mathrm{Quot}_{\mathcal{F}/X/B}(T)$, then $\mathcal{G}_T \to \pi_{T,*}\mathcal{Q}$ is in $\mathrm{Quot}_{\mathcal{G}/Y/B}(T)$. $\square$